

数据与预处理

Data & preprocessing

01

知识速查 / Concepts

核心规则中英对照，数学符号共用。 / Paired explanations; mathematical notation is shared.

学习任务与泛化

Tasks & generalization

聚类：无标签数据分组；分类：预测类别；回归：预测数值；关联规则：发现物品共现。

Clustering groups unlabeled data; classification predicts categories; regression predicts values; association rules discover co-occurrence.

泛化指在未见数据上仍有效；相关不等于因果，训练拟合好不保证泛化好。

Generalization means working on unseen data. Correlation is not causation; good training fit does not guarantee generalization.

属性类型

Attribute types

名义：只有身份，如编号、颜色。有序：有顺序但间隔未必相等，如满意度。

Nominal: identity only, e.g. IDs or colors. Ordinal: ordered levels without necessarily equal gaps, e.g. satisfaction.

区间：差值有意义，无真实零点，如摄氏温度。比率：有真实零点，倍数有意义，如重量。

Interval: meaningful differences but no true zero, e.g. Celsius.

Ratio: true zero and meaningful ratios, e.g. weight.

类型取决于语义，不取决于存储格式。整数 ID 仍是名义属性；20°C 不是 10°C 的两倍热。

Type follows meaning, not storage. An integer ID remains nominal; 20°C is not twice as hot as 10°C.

表示与编码

Representation & encoding

记录：对象×属性；图：节点与边 $G=(V,E)$ ；有序数据：时间序列、空间或顺序信息。

Records: objects × attributes. Graphs: nodes and edges $G=(V,E)$. Ordered data: time, spatial, or sequence information.

独热编码把名义类别变成多个 0/1 特征，避免强加顺序，但会增加维度。

One-hot encoding maps nominal categories to binary features, avoiding an artificial order while increasing dimensionality.

清洗与约简

Cleaning & reduction

清洗处理缺失、重复、错误单位；约简包括抽样、特征选择、降维；聚合可将日期汇总为年份。

Cleaning handles missing values, duplicates, and incorrect units. Reduction includes sampling, feature selection, and dimensionality reduction; aggregation can group dates by year.

分箱平滑可用箱内均值替换值；离散化将连续值转为类别，会丢失细节。

Binning can smooth values using bin means. Discretization converts continuous values into categories and loses detail.

质量与探索

Quality & EDA

噪声是测量或记录扰动；离群点可能是错误，也可能是研究目标；缺失值不能自动当作 0。

Noise is measurement or recording disturbance. Outliers can be errors or the target of analysis. Missing values are not automatically zero.

直方图看分布；箱线图看单变量异常；散点图看关系；同时检查类别比例。

Use histograms for distributions, boxplots for univariate outliers, and scatterplots for relationships; also check class proportions.

异常分布只是线索。单变量正常不代表多变量组合正常；箱线图不能找出所有离群点。

Unusual distributions are clues, not proof. Normal marginal behavior does not imply normal joint behavior; boxplots cannot detect every outlier.

特征缩放

Feature scaling

$$\text{Min-max: } x' = (x - \min) / (\max - \min)$$

$$\text{Z-score: } z = (x - \mu) / \sigma$$

μ 为均值， σ 为标准差；常数特征分母为 0，需单独处理。测试值可以超出训练集 min-max 范围。

μ is the mean and σ the standard deviation. Constant features need special handling because the denominator is zero. Test min-max values can fall outside [0,1].

KNN、K-means 等依赖距离，量纲会影响结果。标准化不会自动使分布变成正态。

Units affect distance-based methods such as KNN and K-means. Standardization does not automatically make data normally distributed.

正确的数据流程

Leakage-free workflow

Split → fit on training data → transform

切分 → 用训练集拟合 → 应用固定变换

仅用训练集估计均值、方差、填补参数与特征选择；验证和测试只应用已拟合变换。

Estimate means, variances, imputation parameters, and feature selection from training data only; apply the fitted transformation to validation and test data.

交叉验证每轮都需独立拟合预处理。最终可在全部开发数据上重拟合，但测试集保持隔离。

Refit preprocessing within each CV training fold. The final pipeline may use all development data while keeping the test set untouched.

数据与预处理

01

Data & preprocessing

例题与解答 / Worked examples

独立编写的教学例题，非历年试题。 / Original teaching examples, not past-exam questions.

E01 属性类型与编码

Attribute types and encoding

题目 / Question

判断属性类型：学生编号、满意度（低/中/高）、摄氏温度、文件大小（字节）。对无序类别“浏览器类型”直接编码为 1、2、3 后用于欧氏距离，有何问题？

Classify these attributes: student ID, satisfaction level (low/medium/high), temperature in degrees Celsius, and file size in bytes. What is wrong with encoding the unordered category “browser type” as 1, 2, and 3 and then using Euclidean distance?

解答 / Solution

学生编号：nominal；满意度：ordinal；摄氏温度：interval；文件大小：ratio。无序类别的数字编码会引入不存在的大小和间距，可用 one-hot。Ordinal 编码也不自动意味着等级之间距离相等。

Student ID: nominal; satisfaction: ordinal; Celsius temperature: interval; file size: ratio. Numeric codes for unordered categories introduce an ordering and distances that do not exist; one-hot encoding is an alternative. Ordinal encoding does not automatically imply equal distances between levels.

易错点 / Watch out

数字外观不等于数值属性；区间尺度的零点可以任意。

Looking like a number does not make an attribute quantitative. An interval scale can have an arbitrary zero point.

E02 标准化与数据泄漏

Standardization and data leakage

题目 / Question

训练集某特征均值 $\mu=20$ 、标准差 $\sigma=5$ 。测试点 $x=32$ 的 z-score 是多少？有人先用全体数据求均值与标准差，再拆分训练/测试集，是否合适？

A training-set feature has mean $\mu=20$ and standard deviation $\sigma=5$. What is the z-score of a test point $x=32$? Is it appropriate to compute the mean and standard deviation using all data before splitting into training and test sets?

解答 / Solution

$z=(32-20)/5=2.4$ 。应先划分数据，再只用训练集拟合标准化参数，固定后变换验证/测试集。交叉验证中也要在每个训练折内部拟合预处理。

$z=(32-20)/5=2.4$. Split the data first, fit standardization parameters using only the training set, then apply those fixed parameters to the validation/test sets. During cross-validation, fit preprocessing separately within each training fold.

易错点 / Watch out

即使没有使用测试标签，使用测试分布拟合预处理仍会泄漏信息。

Fitting preprocessing to the test distribution leaks information even when no test labels are used.

E03 缩放与分布外测试值

Scaling a test value beyond the training range

题目 / Question

训练 $\min=10$ 、 $\max=50$ 、 $\mu=25$ 、 $\sigma=5$ 。求 $x=30$ 的两种缩放结果，以及测试 $x=60$ 的 min-max 值。

Training statistics: $\min=10$, $\max=50$, $\mu=25$, $\sigma=5$. Scale $x=30$ by both methods, then min-max scale test $x=60$.

解答 / Solution

$x=30$ ：min-max= $(30-10)/(50-10)=0.5$ ； $z=(30-25)/5=1$ 。测试 $x=60$ ：min-max= $(60-10)/40=1.25$ 。测试值超出训练范围，因此可大于 1。

For $x=30$, min-max= $(30-10)/40=0.5$ and $z=(30-25)/5=1$. For test $x=60$, min-max= $(60-10)/40=1.25$. A test value outside the training range can exceed 1.

易错点 / Watch out

不要为测试点重算训练 min/max；标准化不保证正态分布。

Do not refit min/max using the test point. Standardization does not guarantee normality.

聚类分析

Clustering

02

知识速查 / Concepts

核心规则中英对照，数学符号共用。 / Paired explanations; mathematical notation is shared.

K-means：目标与迭代

K-means: objective & iteration

$$SSE = \sum_k \sum_{x \in C_k} \|x - \mu_k\|^2$$

$$\mu_k = (1/|C_k|) \sum_{x \in C_k} x$$

初始化 → 分给最近中心 → 更新为簇内均值 → 重复。固定分配时，均值最小化平方误差。

Initialize → assign to nearest centers → update to cluster means → repeat. For fixed assignments, the mean minimizes squared error.

每步 SSE 不增加，但不保证全局最优；对初始化、尺度和离群点敏感，偏好紧凑近球状簇。

SSE does not increase at each step, but global optimality is not guaranteed. Initialization, scale, and outliers matter; compact, roughly spherical clusters are favored.

增大 K 时最优 SSE 不升，不能单凭最小 SSE 选择 K。

Optimal SSE cannot increase as K grows, so minimum SSE alone cannot select K.

变体

Variants

K-means++：按到最近已有中心的距离平方 $D(x)^2$ 抽取新中心；不是总选最远点。

K-means++ samples new centers with probability proportional to $D(x)^2$, the squared distance to the nearest existing center; it does not always pick the farthest point.

K-medoids 选实际样本作中心，通常更抗离群点；X-means 用 BIC 等准则评估拆分、探索 K。

K-medoids uses actual samples as centers and is often more robust to outliers. X-means evaluates splits with criteria such as BIC to explore K.

DBSCAN：点类型与扩展

DBSCAN: points & expansion

ϵ 为邻域半径；MinPts 为最低邻域点数。本表邻域包含自身，距离 $\leq \epsilon$ 算邻居。

ϵ is the neighborhood radius and MinPts the minimum count. Here, neighborhoods include the point itself and use distance $\leq \epsilon$.

核心：邻域数 $\geq$ MinPts；边界：非核心但邻接核心；噪声：两者都不是。

Core: at least MinPts neighbors. Border: non-core but next to a core point. Noise: neither.

只能从核心点扩展；边界点不能搭桥。暂标噪声的点以后可能成为边界点。

Expand only from core points; border points cannot bridge clusters. A temporarily marked noise point may later become a border point.

DBSCAN：适用性

DBSCAN: behavior

不用预设 K，可找非球状簇并排除噪声；不同密度通常难兼顾，可用 k-distance 拐点初选 ϵ 。

No preset K is needed; non-spherical clusters and noise are supported. Varying densities are difficult; a k-distance elbow can guide ϵ .

可能全部为噪声；核心结构固定，但同时邻接两簇的边界点归属可能受访问顺序影响。

All points may be noise. Core structure is fixed for fixed parameters, but a border point adjacent to two clusters may be assigned according to visit order.

AGNES 与连接规则

AGNES & linkage

从一点一簇开始，反复合并最近两簇；在树状图不同高度切开可得到不同簇数。

Start with singleton clusters and repeatedly merge the nearest pair; cutting the dendrogram at different heights gives different clusterings.

Single: $\min d(a,b)$ Complete: $\max d(a,b)$

Average: $\sum d(a,b)/(|A||B|)$

Centroid: $d(\mu_A, \mu_B)$

Single 易形成链；complete 偏紧凑；average 是全部跨簇点对的均值，不是中心距离。

Single linkage can chain; complete linkage favors compact clusters. Average linkage averages all cross-cluster pairs, not center distances.

$$\text{Ward } \Delta SSE = [n_A n_B / (n_A + n_B)] \|\mu_A - \mu_B\|^2$$

Ward 比较合并造成的 SSE 增量。距离矩阵空间 $O(N^2)$ ；朴素 AGNES 时间 $O(N^3)$ 。

Ward compares the increase in SSE on merging. The distance matrix uses $O(N^2)$ space; naive AGNES takes $O(N^3)$ time.

评价

Evaluation

外部指标 purity、Rand index 需要真实标签；内部指标包括 SSE 拐点与 silhouette。

External measures such as purity and Rand index require ground-truth labels; internal measures include SSE elbows and silhouette.

$$s(i) = [b(i) - a(i)] / \max(a(i), b(i))$$

a 为同簇平均距离；b 为到其他各簇平均距离的最小值。近 1 好，近 0 在边界，负值提示错分；单点簇通常记 0。

a is mean within-cluster distance; b is the smallest mean distance to another cluster. Near 1 is good, near 0 indicates a boundary, and negative values suggest misassignment. Singletons usually score 0.

聚类分析

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E04 K-means 一轮更新

One K-means iteration

题目 / Question

一维点 $\{0, 2, 3, 10, 12\}$ ， $k=2$ ，初始中心为 0、10。用欧氏距离分配，更新中心，并计算更新后的 SSE；下一轮分配会变吗？

For the one-dimensional points $\{0, 2, 3, 10, 12\}$, let $k=2$ and initialize the centers at 0 and 10. Assign points using Euclidean distance, update the centers, and compute SSE using the updated centers. Will the assignments change in the next iteration?

解答 / Solution

分簇 $C_1=\{0, 2, 3\}$ ， $C_2=\{10, 12\}$ ；新中心 5/3、11。
 $SSE=(25+1+16)/9+1+1=20/3\approx 6.667$ 。新分界点为
 $(5/3+11)/2=19/3\approx 6.333$ ，分配不变，已达到稳定分配。

The clusters are $C_1=\{0, 2, 3\}$ and $C_2=\{10, 12\}$; the new centers are 5/3 and 11. $SSE=(25+1+16)/9+1+1=20/3\approx 6.667$. The new boundary is $(5/3+11)/2=19/3\approx 6.333$. The assignments remain unchanged, so they are stable.

易错点 / Watch out

SSE 使用平方距离；计算时明确用更新前还是更新后的中心。

SSE uses squared distances. State clearly whether you are using the old or updated centers.

E05 DBSCAN 的三种点

The three types of DBSCAN points

题目 / Question

一维点 $A=0$ 、 $B=0.4$ 、 $C=0.8$ 、 $D=1.6$ 、 $E=4$ 。 $\epsilon=0.9$ ， $\text{MinPts}=3$ ；邻域含自身，距离 $\leq \epsilon$ 算邻居。列出各点邻域大小、点类型、最终簇。

Consider one-dimensional points $A=0$, $B=0.4$, $C=0.8$, $D=1.6$, $E=4$. Let $\epsilon=0.9$ and $\text{MinPts}=3$. Each neighborhood includes the point itself, and distance $\leq \epsilon$ counts as a neighbor. Give each neighborhood size, classify each point, and identify the final clusters.

解答 / Solution

邻域大小 $A:3$ ， $B:3$ ， $C:4$ ， $D:2$ ， $E:1$ 。A/B/C 是 core，D 是 border（在核心点 C 的邻域内），E 是 noise。唯一簇为 $\{A, B, C, D\}$ 。

Neighborhood sizes: $A:3$, $B:3$, $C:4$, $D:2$, $E:1$. A/B/C are core points; D is a border point because it lies in core point C's neighborhood; E is noise. There is one cluster: $\{A, B, C, D\}$.

易错点 / Watch out

边界点不必达到 MinPts；本题已规定计入自身，不要混用另一种约定。

A border point need not satisfy MinPts. This question includes the point itself in its neighborhood; do not switch conventions.

E06 AGNES : single vs complete

AGNES: single versus complete linkage

题目 / Question

一维点 $A=0$ 、 $B=2$ 、 $C=5$ 、 $D=9$ 。分别用 single linkage 和 complete linkage，写出每次合并及其高度。

For one-dimensional points $A=0$, $B=2$, $C=5$, $D=9$, list every merge and its height under single linkage and complete linkage.

解答 / Solution

两者先在距离 2 合并 A、B。Single：距离 3 合并 $\{A, B\}$ 与 C，再在 4 合并 D。Complete： $\{A, B\}$ 到 C 的距离是 5，而 C 到 D 是 4，因此第二步在 4 合并 C、D，最后在 9 合并 $\{A, B\}$ 与 $\{C, D\}$ 。

Both methods first merge A and B at distance 2. Single linkage: merge $\{A, B\}$ with C at 3, then add D at 4. Complete linkage: the distance from $\{A, B\}$ to C is 5, while the distance from C to D is 4; therefore, merge C and D at 4, then merge $\{A, B\}$ with $\{C, D\}$ at 9.

易错点 / Watch out

Single 取跨簇最小距离；complete 取跨簇最大距离。每次合并后重算簇间距离。

Single linkage uses the smallest cross-cluster distance; complete linkage uses the largest. Recompute cluster distances after each merge.

E07 比较连接距离与 Ward

Compare linkage distances and Ward

题目 / Question

$A=\{1, 2\}$ 、 $B=\{6, 8\}$ 。求 single、complete、average，以及 Ward 的 ΔSSE 。

For $A=\{1, 2\}$ and $B=\{6, 8\}$, compute single, complete, average linkage, and Ward's ΔSSE .

解答 / Solution

四个跨簇距离是 5、7、4、6。single=4；complete=7；
 $\text{average}=(5+7+4+6)/4=5.5$ 。中心为 1.5、7，
 $\text{Ward}=(2\times 2/4)\times(7-1.5)^2=30.25$ 。

Cross-cluster distances are 5, 7, 4, 6. Single=4; complete=7; $\text{average}=22/4=5.5$. Centers are 1.5 and 7. $\text{Ward}=(2\times 2/4)\times(7-1.5)^2=30.25$.

易错点 / Watch out

Ward 是 SSE 增量，不能直接当作未平方的中心距离。

Ward is an SSE increase, not the unsquared distance between centers.

关联规则

03

Association rules & Apriori

知识速查 / Concepts

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项集与规则

Itemsets & rules

交易是物品集合；k-项集含 k 个物品。规则 $X \rightarrow Y$ 要求两侧非空且互不相交。

A transaction is a set of items; a k-itemset contains k items. A rule $X \rightarrow Y$ has nonempty, disjoint sides.

$SC(X) = \text{count containing } X$
 $S(X) = SC(X)/N$
 Frequent $\Leftrightarrow S(X) \geq \text{minsup}$

最低支持计数为 $\text{ceil}(N \times \text{minsup})$ 。共现规则不表示因果。

The minimum count is $\text{ceil}(N \times \text{minsup})$. Co-occurrence rules do not imply causation.

支持度、置信度、提升度

Support, confidence & lift

$\text{Support}(X \rightarrow Y) = S(X \cup Y)$
 $\text{Confidence}(X \rightarrow Y) = S(X \cup Y)/S(X)$
 $\text{Lift}(X \rightarrow Y) = S(X \cup Y)/[S(X)S(Y)]$

支持度：同时出现的比例；置信度：有 X 时出现 Y 的概率；提升度：相对于 Y 基础概率的倍数。

Support is the fraction containing both sides; confidence is $P(Y|X)$; lift is confidence relative to the base probability of Y.

Lift > 1 为正关联， $= 1$ 对该对事件满足独立关系， < 1 为负关联。高置信度未必高 lift。

Lift > 1 means positive association, $= 1$ independence for the two events, and < 1 negative association. High confidence need not imply high lift.

支持度和 lift 对换方向不变；置信度一般有方向。 $S(X)=0$ 时置信度未定义。

Support and lift are symmetric; confidence generally is not. Confidence is undefined when $S(X)=0$.

搜索规模

Search space

d items: $2^d - 1$ nonempty itemsets
 All rules: $3^d - 2^{d+1} + 1$
 From k items: $2^k - 2$ rules

每条规则的物品可属于前件、后件或都不属于，再排除空前件或空后件。

Each item belongs to the antecedent, consequent, or neither; exclude rules with an empty side.

Apriori 原理与流程

Apriori principle & workflow

$A \subseteq B \Rightarrow S(A) \geq S(B)$

频繁项集的全部子集都频繁；不频繁项集的全部超集都不频繁。子集都频繁不保证超集频繁。

Every subset of a frequent itemset is frequent; every superset of an infrequent set is infrequent. Frequent subsets do not guarantee a frequent superset.

连接上一层频繁项集 $\rightarrow$ 若任一直接子集不频繁则剪枝 $\rightarrow$ 扫描计数 $\rightarrow$ 按 minsup 过滤。

Join frequent itemsets from the previous level $\rightarrow$ prune if any immediate subset is infrequent $\rightarrow$ scan and count $\rightarrow$ filter by minsup.

讲义使用 F_k 表示频繁集、 L_k 表示候选集；部分练习用 L_k 表示频繁集，先看定义。

The lectures use F_k for frequent sets and L_k for candidates. Some exercises use L_k for frequent sets; check the stated convention.

规则生成与剪枝

Rule generation & pruning

对频繁集 S 作非空二分 $X \rightarrow S \setminus X$ ，复用支持计数，检查 minconf；规则须同时通过支持度和置信度门槛。

Split a frequent set S into $X \rightarrow S \setminus X$, reuse support counts, and check minconf. A rule must pass both support and confidence thresholds.

固定并集 S，扩大后件会缩小前件，分母支持度不减，因此置信度不增。

For fixed union S, enlarging the consequent shrinks the antecedent. Its support cannot decrease, so confidence cannot increase.

若 $ABC \rightarrow D$ 不过门槛，则 $AB \rightarrow CD$ 、 $A \rightarrow BCD$ 也不过；不同并集的规则不能直接这样比较。Lift 不适用支持度剪枝。

If $ABC \rightarrow D$ fails, $AB \rightarrow CD$ and $A \rightarrow BCD$ also fail for the same union. Do not apply this to arbitrary unions or apply support pruning to lift.

闭合与最大频繁集

Closed & maximal frequent sets

闭合：没有支持度相同的真超集；最大频繁：没有频繁真超集。最大频繁集必为闭合频繁集，反之不一定。

Closed: no proper superset has the same support. Maximal frequent: no proper superset is frequent. Maximal frequent sets are closed, but not conversely.

关联规则

03

Association rules & Apriori

例题与解答 / Worked examples

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E08 Support / confidence / lift

Support, confidence, and lift

题目 / Question

200 笔交易中，A 出现 80 次，B 出现 150 次，A、B 同时出现 50 次。计算 $A \rightarrow B$ 的 support、confidence、lift，以及 $B \rightarrow A$ 的 confidence。

Among 200 transactions, A appears in 80, B in 150, and both A and B in 50. Compute support, confidence, and lift for $A \rightarrow B$, and confidence for $B \rightarrow A$.

解答 / Solution

support=50/200=0.25; confidence($A \rightarrow B$)=50/80=0.625; lift=0.625/(150/200)=5/6 $\approx$ 0.833; confidence($B \rightarrow A$)=50/150=1/3。Lift<1，说明本数据中 A 的出现与 B 呈负向关联，不能由 62.5% 的置信度断言正向关联。

Support=50/200=0.25; confidence($A \rightarrow B$)=50/80=0.625; lift=0.625/(150/200)=5/6 $\approx$ 0.833; confidence($B \rightarrow A$)=50/150=1/3. Lift<1 indicates a negative association between A and B in these data. A confidence of 62.5% alone does not establish a positive association.

易错点 / Watch out

规则的 support 是交集占全部交易的比例；confidence 的分母是前件。

Rule support is the fraction of all transactions containing both items. The confidence denominator is the antecedent count.

E09 Apriori 候选剪枝

Apriori candidate pruning

题目 / Question

频繁二项集 $L_2 = \{AB, AC, AD, BC, BD\}$ ，CD 不频繁。按字母序连接与剪枝，哪些三项候选保留？是否可以直接说它们频繁？

The frequent 2-itemsets are $L_2 = \{AB, AC, AD, BC, BD\}$; CD is not frequent. Using lexicographic joining and pruning, which 3-item candidates survive? Can you immediately conclude that they are frequent?

解答 / Solution

连接可产生 ABC、ABD、ACD；ACD 因 CD 不频繁被剪掉。最终保留 ABC、ABD。BCD 同样因 CD 不频繁不可能频繁。保留下来的候选还必须扫描交易、计算支持度，才能确定是否频繁。

Joining produces ABC, ABD, and ACD. Prune ACD because CD is not frequent, leaving ABC and ABD. BCD also cannot be frequent because CD is not frequent. The surviving candidates still require a transaction scan and support counting to determine whether they are frequent.

易错点 / Watch out

所有子集频繁只是必要条件，不是充分条件。

Having only frequent subsets is a necessary condition, not a sufficient condition.

E10 Closed 与 maximal

Closed and maximal frequent itemsets

题目 / Question

仅有项目 A、B、C。交易为 ABC、ABC、AB、A，minsup=50%。列出所有非空频繁项集、闭合频繁项集和最大频繁项集。

The only items are A, B, and C. The transactions are ABC, ABC, AB, and A, with minsup=50%. List all nonempty frequent itemsets, all closed frequent itemsets, and all maximal frequent itemsets.

解答 / Solution

支持计数：A=4，B=3，C=2，AB=3，AC=2，BC=2，ABC=2；七个都频繁。Closed：A、AB、ABC；maximal：仅 ABC。B 与其真超集 AB 支持度相同，故不 closed；AC、BC、C 与 ABC 支持度相同。

Support counts: A=4, B=3, C=2, AB=3, AC=2, BC=2, ABC=2; all seven are frequent. Closed: A, AB, ABC. Maximal: ABC only. B is not closed because its proper superset AB has the same support; AC, BC, and C have the same support as ABC.

易错点 / Watch out

Closed：没有相同支持度的真超集。Maximal：没有任何频繁真超集。

Closed: no proper superset has the same support. Maximal frequent: no proper superset is frequent.

E11 固定并集的规则剪枝

Confidence pruning with a fixed union

题目 / Question

100 笔交易中，SC(ABCD)=12、SC(ABC)=20、SC(AB)=40、SC(A)=60。minconf=0.7，判断 $ABC \rightarrow D$ 、 $AB \rightarrow CD$ 、 $A \rightarrow BCD$ 。

Among 100 transactions, SC(ABCD)=12, SC(ABC)=20, SC(AB)=40, SC(A)=60. With minconf=0.7, evaluate $ABC \rightarrow D$, $AB \rightarrow CD$, and $A \rightarrow BCD$.

解答 / Solution

三个并集都为 ABCD，支持度均为 12/100=0.12。置信度依次为 12/20=0.6、12/40=0.3、12/60=0.2，全部未达门槛。 $ABC \rightarrow D$ 已失败，后两条可按固定并集规则剪枝。

All unions are ABCD, giving support=0.12. Confidences are 12/20=0.6, 12/40=0.3, and 12/60=0.2; all fail. Once $ABC \rightarrow D$ fails, the other two may be pruned using the fixed-union rule.

易错点 / Watch out

并集固定时才有这个方向的置信度单调性；minconf 与 minsup 不能混用。

This confidence monotonicity requires a fixed union. Do not confuse minconf with minsup.

分类与决策树

Classification, KNN & decision trees

知识速查 / Concepts

核心规则中英对照，数学符号共用。 / Paired explanations; mathematical notation is shared.

混淆矩阵与指标

Confusion matrix & metrics

先确认正类和坐标轴。TP 真正例，FP 误报，FN 漏报，TN 真负例；实际正类=TP+FN。

Check the positive class and axes first. TP: true positive; FP: false alarm; FN: miss; TN: true negative. Actual positives=TP+FN.

Accuracy = (TP+TN)/N
Precision = TP/(TP+FP)
Recall = TP/(TP+FN)
Specificity = TN/(TN+FP)
F1 = 2PR/(P+R) = 2TP/(2TP+FP+FN)

Precision 看预测正类有多少是真的；recall 看实际正类找回多少。F1 是调和平均；分母为 0 要按约定处理。

Precision measures correctness among positive predictions; recall measures coverage of actual positives. F1 is a harmonic mean; zero denominators need a convention.

类别失衡时 accuracy 可能误导；总猜多数类也能有高准确率。

Accuracy can mislead under class imbalance; always predicting the majority class may score highly.

多分类平均

Multiclass averaging

One-vs-rest 逐类计算。Micro 先汇总计数再算；macro 先逐类算再等权平均。

Use one-vs-rest per class. Micro pools counts before computing; macro computes per-class metrics and averages equally.

单标签、包含全部类别时，micro-P=micro-R=micro-F1=accuracy。Macro-F1 是各类 F1 的平均。

For single-label tasks including all classes, micro-P=micro-R=micro-F1=accuracy. Macro-F1 is the mean of per-class F1, not F1 of macro-P/R.

KNN

Nearest neighbors

分类多数投票，回归取目标均值。小 k 高方差，大 k 更平滑且可能欠拟合；距离对尺度敏感。

Classify by majority vote and regress by averaging targets. Small k has high variance; large k smooths and may underfit. Distances depend on feature scales.

高维距离可能趋同。奇数 k 不能保证多分类无平票；KNN 的 k 是邻居数。

Distances can become less informative in high dimensions. Odd k does not prevent multiclass ties; KNN's k counts neighbors.

贪心分裂

Greedy splits

每步选择局部最好分裂，不保证全局最优。分类叶预测多数类；回归叶预测均值。

Each step chooses the best local split without guaranteeing a globally optimal tree. Classification leaves predict the majority; regression leaves predict the mean.

Gini = $1 - \sum c p(c)^2$
Entropy = $-\sum c p(c) \log_2 p(c)$
I_split = $\sum j (N_j/N) I(\text{child } j)$
Gain = I(parent) - I_split

纯节点不纯度为 0，约定 $0 \log 0 = 0$ 。子节点按样本数加权，不是简单平均。

Pure nodes have zero impurity; define $0 \log 0 = 0$. Weight child impurities by sample counts rather than taking a simple mean.

候选与回归

Candidates & regression

数值：排序后在相邻不同值间试阈值。有序 d 类：d-1 个切点。

名义 d 类： $2^{(d-1)}-1$ 个无序非空二分。

Numeric: try thresholds between adjacent distinct sorted values.

Ordered d levels: d-1 cuts. Nominal d categories: $2^{(d-1)}-1$ unordered nonempty binary partitions.

RSS_split = $\sum j \sum_{i \in \text{child } j} (y_i - \text{mean}_j)^2$

回归选子节点总 RSS 最小的分裂；不能直接相加已经除以样本数的方差。

Choose the smallest total child RSS for regression; do not simply add variances that have already been divided by child sizes.

过拟合与剪枝

Overfitting & pruning

深树高方差、易记忆训练数据。预剪枝限制深度或节点样本数；后剪枝先建树，再根据验证准则删分裂。

Deep trees have high variance and can memorize training data. Pre-pruning limits depth or node sizes; post-pruning removes splits after fitting using validation criteria.

根深度为 0 时，N 个可区分点最大深度可达 N-1；完全相同特征但标签冲突时不能完美拟合。

With root depth 0, N distinguishable points can yield depth N-1.

Identical features with conflicting labels prevent perfect fitting.

分类与决策树

Classification, KNN & decision trees

例题与解答 / Worked examples

独立编写的教学例题，非历年试题。 / Original teaching examples, not past-exam questions.

E12 混淆矩阵与 F1

Confusion matrix and F1

题目 / Question

以“违约”为正类：TP=36, FP=12, FN=24, TN=128。计算 accuracy、precision、recall、F1；总预测为负的基线模型 accuracy 是多少？

Treat “default” as the positive class: TP=36, FP=12, FN=24, TN=128. Compute accuracy, precision, recall, and F1. What accuracy would a baseline that always predicts negative achieve?

解答 / Solution

总数 200。Accuracy=164/200=0.82；precision=36/48=0.75；recall=36/60=0.60；F1=2TP/(2TP+FP+FN)=72/108=2/3≈0.667。全预测负：accuracy=140/200=0.70，但正类 recall=0。

There are 200 examples. Accuracy=164/200=0.82; precision=36/48=0.75; recall=36/60=0.60; F1=2TP/(2TP+FP+FN)=72/108=2/3≈0.667. Always predicting negative gives accuracy=140/200=0.70, but positive-class recall=0.

易错点 / Watch out

TN 不是实际负类总数；实际负类=TN+FP。

TN is not the total number of actual negatives; actual negatives=TN+FP.

E13 决策树分裂

Decision-tree splits

题目 / Question

父节点有 4 正 4 负。分裂 A：左 3 正 1 负，右 1 正 3 负；分裂 B：左 2 正 0 负，右 2 正 4 负。分别算信息增益与 Gini 降低量，选哪个？log 以 2 为底。

A parent node contains 4 positives and 4 negatives. Split A: left child has 3 positives and 1 negative; right child has 1 positive and 3 negatives. Split B: left has 2 positives and 0 negatives; right has 2 positives and 4 negatives. Compute information gain and Gini decrease for both splits. Which split should be chosen? Use base-2 logarithms.

解答 / Solution

父熵=1。A：加权子熵 $H(3/4)=0.8113$ ，IG=0.1887。B：子熵 $= (2/8) \times 0 + (6/8) \times H(1/3) = 0.6887$ ，IG=0.3113，选 B。父 Gini=0.5；A 子 Gini=0.375，降低 0.125；B 子 Gini= $(6/8) \times 4/9 = 1/3$ ，降低 1/6，仍选 B。

Parent entropy=1. A: weighted child entropy $H(3/4)=0.8113$, so IG=0.1887. B: weighted child entropy $= (2/8) \times 0 + (6/8) \times H(1/3) = 0.6887$, so IG=0.3113; choose B. Parent Gini=0.5. A: child Gini=0.375, decrease=0.125. B: weighted child Gini= $(6/8) \times 4/9 = 1/3$, decrease=1/6; again choose B.

易错点 / Watch out

对子节点不纯度按样本数加权，不能直接求简单平均。

Weight child impurities by their sample counts; do not take an unweighted average.

E14 KNN 与尺度

KNN and feature scales

题目 / Question

到查询点最近的三条记录依次为：距离 0.2 标签 +；距离 0.5 标签 -；距离 0.8 标签 -。用不加权投票预测 k=1 与 k=3。若特征中收入按元、年龄按年计，为什么应考虑标准化？

The three nearest records to a query point are, in order: distance 0.2, label +; distance 0.5, label -; distance 0.8, label -. Predict using unweighted voting for k=1 and k=3. If income is measured in currency units and age in years, why consider standardization?

解答 / Solution

1-NN 为 +；3-NN 为 -。距离投票只按数量，不按距离加权。量纲和数值范围会改变欧氏距离各项的贡献，收入可能支配距离；可只用训练集拟合标准化参数。

1-NN predicts +; 3-NN predicts -. Unweighted voting counts neighbors without distance weights. Units and numerical ranges change each feature's contribution to Euclidean distance, so income may dominate. Fit standardization parameters using only the training set.

易错点 / Watch out

题目若没有说加权投票，不要自行使用 1/d 权重。

Do not introduce 1/d weights unless the question specifies weighted voting.

回归与模型验证

Regression & validation

知识速查 / Concepts

核心规则中英对照，数学符号共用。 / Paired explanations; mathematical notation is shared.

线性回归

Linear regression

$$\hat{y} = x^T \theta \quad \text{MSE} = \|X\theta - y\|^2/n$$

$$\nabla \text{MSE} = 2X^T(X\theta - y)/n$$

$$\theta = (X^T X)^{-1} X^T y; \text{ generally } \theta = X^+ y$$

含截距时 $x_0=1$ 。系数表示其他变量固定时的预测变化，不自动代表因果。

With an intercept, $x_0=1$. A coefficient describes predicted change with other variables fixed; it is not automatically causal.

求逆式要求满列秩；共线或参数过多会奇异，伪逆给最小范数最小二乘解。

The inverse formula requires full column rank. Collinearity or too many parameters can cause singularity; the pseudoinverse gives a minimum-norm least-squares solution.

梯度下降

Gradient descent

$$\theta_{\text{new}} = \theta_{\text{old}} - \eta \nabla L(\theta_{\text{old}})$$

同一组旧参数算全部梯度后同步更新。学习率太大可能发散，太小收敛慢。

Compute every gradient component at the same old parameters, then update simultaneously. Too large a learning rate may diverge; too small is slow.

Batch 用全部样本；SGD 用单样本；mini-batch 用一小批。损失若用 $1/(2n)$ ，MSE 梯度系数减半。

Batch uses all samples; SGD one; mini-batch a subset. Using $1/(2n)$ in squared loss halves the MSE gradient.

多项式与 L2 正则

Polynomial features & L2

多项式仍对参数线性； d 个变量最高 p 次、含常数项共有 $C(d+p, p)$ 项。

Polynomial regression remains linear in its parameters. With d variables and degree at most p , there are $C(d+p, p)$ terms including the constant.

$$L = \|X\theta - y\|^2/n + (\lambda/n) \theta^T D \theta$$

$$D = \text{diag}(0, 1, \dots, 1)$$

$$\theta = (X^T X + \lambda D)^{-1} X^T y$$

$$\nabla L = 2X^T(X\theta - y)/n + 2\lambda D\theta/n$$

通常不罚截距； λ 越大约束越强，可欠拟合。尺度影响惩罚，正则不保证降低训练误差。

Usually the intercept is unpenalized. Larger λ constrains more and may underfit. Scaling affects penalties; regularization need not lower training error.

逻辑回归

Logistic regression

$$z = x^T \theta; p = \sigma(z) = 1/(1+e^{-z})$$

$$L = -\sum_i [y_i \ln p_i + (1-y_i) \ln(1-p_i)]/n$$

$$\nabla L = X^T(p - y)/n$$

$y \in \{0, 1\}$, $p = P(y=1|x)$ 。二元交叉熵来自负对数似然，通常迭代优化，没有普通正规方程。

$y \in \{0, 1\}$ and $p = P(y=1|x)$. Binary cross-entropy is negative log-likelihood and is usually optimized iteratively, without an ordinary normal equation.

$p \geq \tau$ 判为 1； $\tau=0.5$ 时边界 $z=0$ 。多项式特征可产生原空间非线性边界。

Predict 1 when $p \geq \tau$; at $\tau=0.5$ the boundary is $z=0$. Polynomial features can give nonlinear boundaries in the original space.

改阈值不改概率或交叉熵。提高阈值会减少预测正例；分对也可能有非零损失。

Changing the threshold does not change probabilities or cross-entropy. Raising it reduces positive predictions; a correct decision can still have nonzero loss.

凸性与数值条件

Convexity & conditioning

OLS 和标准 logistic loss 对参数凸，但不一定有唯一有限最优；可分数据的无正则 logistic 参数可不断增大。

OLS and standard logistic loss are convex in parameters, but need not have a unique finite optimum; unregularized logistic coefficients can diverge on separable data.

可逆仿射缩放可保持无正则线性模型表达能力、改变系数，并改善梯度下降数值条件。

Invertible affine scaling can preserve an unregularized linear model's expressiveness, change coefficients, and improve gradient-descent conditioning.

验证与泄漏

Validation & leakage

训练拟合参数；验证选择模型和超参数；测试只作最终评价。

Training fits parameters; validation selects models and hyperparameters; testing gives the final evaluation.

K 折 CV 每轮用 $K-1$ 折训练、1 折验证；每轮仅用训练折拟合填补、缩放等预处理。

K-fold CV trains on $K-1$ folds and validates on the remaining fold; fit imputation, scaling, and other preprocessing only within that training fold.

不能看测试分数反复改模型，否则测试集实际变成验证集。

Repeatedly changing models based on test scores turns the test set into a validation set.

回归与模型验证

Regression & validation

例题与解答 / Worked examples

独立编写的教学例题，非历年试题。 / Original teaching examples, not past-exam questions.

E15 线性回归梯度

Linear-regression gradients

题目 / Question

数据 $(x,y)=\{(1,2),(2,4)\}$ ，模型 $\hat{y}=wx+b$ ， $J=(1/2n)\sum(\hat{y}-y)^2$ 。初始 $w=b=0$ ， $\eta=0.1$ ；同步更新一次 w 、 b ，并求更新后的 J 。

Given $(x,y)=\{(1,2),(2,4)\}$, model $\hat{y}=wx+b$, and loss $J=(1/(2n))\sum(\hat{y}-y)^2$, initialize $w=b=0$ and use $\eta=0.1$. Update w and b simultaneously for one step, then compute the new J .

解答 / Solution

$\partial J/\partial w=(-2\times 1-4\times 2)/2=-5$ ； $\partial J/\partial b=(-2-4)/2=-3$ 。 $w_{\text{新}}=0.5$ ， $b_{\text{新}}=0.3$ 。新预测 0.8、1.3； $J=(1.44+7.29)/4=2.1825$ 。

$\partial J/\partial w=(-2\times 1-4\times 2)/2=-5$ ； $\partial J/\partial b=(-2-4)/2=-3$ 。The updated parameters are $w=0.5$ and $b=0.3$. The new predictions are 0.8 and 1.3, so $J=(1.44+7.29)/4=2.1825$.

易错点 / Watch out

必须根据同一组旧参数求全部梯度；损失中的 $1/2n$ 决定是否出现系数 2。

Compute all gradients from the same old parameters. The $1/(2n)$ factor in the loss determines whether a factor of 2 remains in the gradient.

E16 Logistic：概率、损失与梯度

Logistic regression: probability, loss, and gradients

题目 / Question

单样本 $x=2$ 、 $y=1$ ，模型 $p=\sigma(wx+b)$ ，初始 $w=0.5$ 、 $b=-1$ 。求 p 、二元交叉熵（自然对数），阈值 0.6 下的预测；再用 $\eta=0.1$ 更新一次。

For one example $x=2$, $y=1$, use $p=\sigma(wx+b)$ with initial $w=0.5$, $b=-1$. Compute p , binary cross-entropy using natural logarithms, and the prediction at threshold 0.6. Then take one update step with $\eta=0.1$.

解答 / Solution

$z=0$ ， $p=0.5$ ；loss $=-\ln(0.5)=0.6931$ 。阈值 0.6 下预测为 0。
 $\partial L/\partial w=(p-y)x=-1$ ， $\partial L/\partial b=p-y=-0.5$ ，因此 $w_{\text{新}}=0.6$ ， $b_{\text{新}}=-0.95$ 。新 $z=0.25$ ， $p\approx 0.5622$ ，仍未达到阈值。

$z=0$ and $p=0.5$; loss $=-\ln(0.5)=0.6931$. The prediction at threshold 0.6 is 0. $\partial L/\partial w=(p-y)x=-1$ and $\partial L/\partial b=p-y=-0.5$, giving updated $w=0.6$ and $b=-0.95$. The new $z=0.25$ and $p\approx 0.5622$, still below the threshold.

易错点 / Watch out

概率、离散预测和损失是三件事；损失不依赖你选择的分类阈值。

Probability, discrete prediction, and loss are different quantities. The loss does not depend on the classification threshold you choose.

E17 模型选择与测试集

Model selection and the test set

题目 / Question

三模型 A/B/C 的平均验证准确率为 0.79/0.83/0.81；测试准确率为 0.88/0.82/0.85。应依据哪一列选模型？5-fold CV 时能否先在全部五折上拟合缺失值填补器？

Models A/B/C have mean validation accuracies 0.79/0.83/0.81 and test accuracies 0.88/0.82/0.85. Which scores should determine model selection? In 5-fold cross-validation, may you fit a missing-value imputer on all five folds before training?

解答 / Solution

按验证结果选择 B。测试集只在最终方案确定后用于评估，不能根据测试分数改选 A。每个 CV 训练折内部独立拟合填补器及其他预处理，再应用于该轮验证折。最终可在全部开发数据上重新拟合已选方案。

Select B using validation results. Use the test set only to evaluate the finalized choice; do not switch to A because of its test score. Fit the imputer and all other preprocessing independently on each CV training fold, then apply them to that iteration's validation fold. Finally, the chosen pipeline can be refitted on all development data.

易错点 / Watch out

多次查看测试成绩并据此改模型，会让测试集变成验证集。

Repeatedly inspecting test scores and changing the model accordingly turns the test set into a validation set.

集成学习

06

Ensemble learning

知识速查 / Concepts

核心规则中英对照，数学符号共用。 / Paired explanations; mathematical notation is shared.

组合、Bagging 与随机森林

Combination, bagging & RF

投票/平均组合独立模型；stacking 用元模型学习组合，须用无泄漏的基模型预测训练。

Voting/averaging combines independently trained models. Stacking learns a meta-model from base predictions that must be generated without leakage.

Bagging 从 N 个样本有放回抽 N 次，独立训练可并行，分类投票、回归平均，主要降低方差。

Bagging draws N times with replacement from N samples and trains independently in parallel. Classification votes; regression averages; variance is mainly reduced.

随机森林额外在每个节点分裂时随机选候选特征，降低树间相关性；不是每树固定一套特征。

RF also samples candidate features at each node split to reduce tree correlation; it does not simply fix one feature subset per tree.

$$P(\text{OOB for one tree}) = (1-1/N)^N \approx e^{-1} \approx .368$$

OOB 预测只用抽样时未见该记录的树。更多树通常不单独增加过拟合，但增加成本。

OOB prediction uses only trees whose samples excluded the record. More trees usually do not by themselves increase overfitting, but cost more.

AdaBoost

Adaptive boosting

Initialize $w_i = 1/N$; labels $y_i \in \{-1, +1\}$
 $\epsilon_m = \sum_i w_i I[h_m(x_i) \neq y_i]$
 $\alpha_m = \frac{1}{2} \ln[(1-\epsilon_m)/\epsilon_m]$
 $w_i \leftarrow w_i \exp(-\alpha_m y_i h_m(x_i))$; normalize
 Prediction = $\text{sign}(\sum_m \alpha_m h_m(x))$

ϵ 是加权错误率，不是错误个数。错例乘 e^α ，对例乘 $e^{-\alpha}$ ，再统一归一化。

ϵ is weighted error, not a raw count. Multiply wrong examples by e^α and correct ones by $e^{-\alpha}$, then normalize.

$\epsilon=0.5$ 时 $\alpha=0$ ； $\epsilon>0.5$ 需处理； $\epsilon=0$ 时停止或数值保护。噪声也可能被不断增权。

At $\epsilon=0.5$, $\alpha=0$; $\epsilon>0.5$ needs handling; at $\epsilon=0$ stop or use numerical safeguards. Noise can also receive increasing weight.

梯度提升

Gradient boosting

Squared-error regression:

$$F_0(x) = \text{mean}(y)$$

$$r_i = y_i - F_{m-1}(x_i)$$

$$\text{Fit } h_m \text{ to } r_i; F_m = F_{m-1} + \eta h_m$$

残差是真值减当前预测；更新加的是树预测的残差，不是真实残差。

A residual is target minus current prediction. The update adds the tree's predicted residual, not the true residual.

一般形式拟合损失负梯度；分类常更新分数，再用 sigmoid/softmax 变成概率。

The general form fits negative loss gradients. Classification often updates scores, then maps them to probabilities using sigmoid/softmax.

Bagging 独立并行，boosting 前后依赖；boosting 常降低偏差，但也可能过拟合。

Bagging trains independently in parallel; boosting rounds depend on earlier rounds. Boosting often reduces bias but can overfit.

XGBoost

Regularized boosted trees

$$\text{Objective} = \sum_i \ell(y_i, \hat{y}_i) + \sum_m \Omega(f_m)$$

$$\Omega(f) = \gamma T + (\lambda/2) \sum_j w_j^2$$

利用一、二阶导数近似损失； T 是叶数、 w 是叶值而非样本权重。 γ 抑制叶数， λ 抑制极端叶值。

Uses first- and second-order loss derivatives. T counts leaves and w denotes leaf values, not sample weights. γ penalizes leaves and λ extreme leaf values.

还支持行列采样、稀疏处理与树内部并行。

Also supports row/column sampling, sparse handling, and parallelism within tree construction.

LightGBM 与 CatBoost

Other boosted-tree methods

LightGBM：直方图分裂，leaf-wise 优先拆收益最大叶；GOSS 留大梯度、抽小梯度并重加权；EFB 合并互斥稀疏特征。

LightGBM uses histogram splits and leaf-wise growth. GOSS keeps large gradients and samples/reweights small ones; EFB bundles mutually exclusive sparse features.

CatBoost：有序目标编码仅用排列中此前同类样本并平滑；ordered boosting 减少预测偏移；对称树同层用相同分裂。

CatBoost's ordered target encoding uses earlier same-category samples with smoothing. Ordered boosting reduces prediction shift; symmetric trees share a split at each depth.

独立编写的教学例题，非历年试题。 / Original teaching examples, not past-exam questions.

E18 Bootstrap 与 OOB

Bootstrap and out-of-bag samples

题目 / Question

从 n 条训练记录中有放回抽 n 次。某记录从未被抽到的概率是多少？当 n 很大时约多少？随机森林除了 bootstrap 还引入什么随机性，目的是什么？

Draw n times with replacement from n training records. What is the probability that a particular record is never drawn, and its approximation for large n ? What additional randomness does a random forest introduce besides bootstrap sampling, and why?

解答 / Solution

未抽到概率 $= (1 - 1/n)^n$ ， n 大时约 $e^{-1} \approx 0.368$ 。随机森林在节点分裂时随机选候选特征子集，通常能降低树之间的相关性，使集成更有效地降低方差。OOB 预测只使用没见过该样本的树。

The probability is $(1 - 1/n)^n$, approaching $e^{-1} \approx 0.368$ for large n . At each node split, a random forest selects a random subset of candidate features. This usually reduces correlation between trees, allowing the ensemble to reduce variance more effectively. An OOB prediction uses only trees whose bootstrap samples excluded that record.

易错点 / Watch out

Bootstrap 样本大小是 n ，但通常不含 n 个不同样本；OOB 不是额外收集的测试集。

A bootstrap sample contains n draws but usually fewer than n distinct records. OOB data are not an additionally collected test set.

E19 AdaBoost 一轮权重

One AdaBoost weight update

题目 / Question

4 个样本初始权重各 $1/4$ ，弱分类器只错第 1 个。标签及预测均为 ± 1 ； $\alpha = \frac{1}{2} \ln((1 - \epsilon)/\epsilon)$ ， $D_{\text{new}}(i) \propto D(i) \exp(-\alpha y_i h_i)$ 。求 ϵ 、 α 和归一化后的权重。

Four examples initially have weight $1/4$ each. A weak classifier misclassifies only the first example. Labels and predictions are ± 1 ； $\alpha = \frac{1}{2} \ln((1 - \epsilon)/\epsilon)$, and $D_{\text{new}}(i) \propto D(i) \exp(-\alpha y_i h_i)$. Find ϵ , α , and the normalized updated weights.

解答 / Solution

$\epsilon = 1/4$ ； $\alpha = \frac{1}{2} \ln 3 \approx 0.5493$ 。错例乘 $e^\alpha = \sqrt{3}$ ，对例乘 $e^{-\alpha} = 1/\sqrt{3}$ 。归一化常数 $Z = \sqrt{3}/4 + 3/(4\sqrt{3}) = \sqrt{3}/2$ 。新权重为 $(1/2, 1/6, 1/6, 1/6)$ 。

$\epsilon = 1/4$ ； $\alpha = \frac{1}{2} \ln 3 \approx 0.5493$. Multiply the misclassified example's weight by $e^\alpha = \sqrt{3}$ and each correct example's weight by $e^{-\alpha} = 1/\sqrt{3}$. The normalization constant is $Z = \sqrt{3}/4 + 3/(4\sqrt{3}) = \sqrt{3}/2$. The new weights are $(1/2, 1/6, 1/6, 1/6)$.

易错点 / Watch out

先乘权重因子，再对全体样本归一化；错例总权重增大，但并非简单加一个常数。

Apply the multiplicative factors first, then normalize over all examples. The total weight of misclassified examples increases; this is not an additive constant update.

E20 梯度提升的一步更新

One gradient-boosting update

题目 / Question

$y=1400$ ，当前 $F=1000$ ，新树预测残差 $h=350$ ，学习率 $\eta=0.1$ 。求原残差、新预测、新残差。

Given $y=1400$, current $F=1000$, new tree residual prediction $h=350$, and $\eta=0.1$, find the old residual, updated prediction, and new residual.

解答 / Solution

原残差 $r=1400-1000=400$ 。新预测 $F=1000+0.1 \times 350=1035$ 。新残差 $=1400-1035=365$ 。更新用树预测的 350，不是用真实残差 400。

Old residual $=400$. Updated prediction $=1000+0.1 \times 350=1035$. New residual $=1400-1035=365$. The update uses the tree's prediction 350, not the true residual 400.

易错点 / Watch out

残差方向是真值减预测；学习率乘在新树贡献上。

Residual means target minus prediction. Apply the learning rate to the new tree's contribution.